

Week 2 Progress Report

Literature Review (continued) and Dataset Collection

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

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Course: Open Data Practice - Research Practicum

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Gantt activities this week (W2): Literature Review (W1-W3), Dataset Collection (W2-W3)

Section	Content
1	Week 2 scope against the project Gantt chart
2	Literature review: thematic synthesis, method comparison matrix, refined gaps
3	Dataset collection: selection, acquisition, data dictionary, quality audit, first look
4	Ethics and data governance
5	Progress status, risks and Week 3 plan

1. Week 2 Scope

The project Gantt chart (CA2 proposal, Section 3.7, Figure 1) plans twelve weeks. Literature Review runs W1-W3 and Dataset Collection runs W2-W3; Data Preprocessing starts in W3 and Exploratory Data Analysis in W4. Week 2 therefore has two tasks: turn the Week 1 paper summaries into a critical synthesis, and obtain, verify and document the dataset. Full EDA is left for W4 as planned; Section 3.6 gives only the first-look checks needed to plan preprocessing.

Gantt activity	Planned weeks	Week 2 target	Status
Literature Review	W1-W3	Thematic synthesis, comparison matrix, refined gaps	On track
Dataset Collection	W2-W3	Dataset selected, downloaded, verified and documented	Ahead (core done)
Data Preprocessing	W3-W4	Not started; issues logged in Section 3.6 feed it	Next week
Exploratory Data Analysis	W4-W5	First-look checks only	Planned

2. Literature Review (continued)

2.1 Thematic synthesis

Week 1 summarised 15 papers (2023-2026) one at a time. This week they were compared with each other. Five findings shape the study design.

(a) Random Forest is the consistent leader, Logistic Regression the recall baseline. Random Forest gives the best F1 in Papers 4, 5, 7, 8, 13 and 15, and the reviews (Papers 3, 9, 14) report the same pattern: median F1 of about 0.82 for ensembles against 0.65 for single classifiers (Paper 3). Logistic Regression often has the highest recall but low precision (Papers 4, 6). This supports the proposal's comparison of Logistic Regression, Decision Tree and Random Forest.

(b) SMOTE trades precision for recall, and the timing matters. SMOTE raised Random Forest F1 from 0.82 to 0.91 but lowered precision from 0.93 to 0.87 (Paper 7), and raised recall from 0.76 to 0.92 (Paper 8). Most papers do not say whether SMOTE was applied before or after the train-test split. Applying it before the split leaks synthetic copies of test frauds into training and inflates scores. This study will resample the training fold only and compare against class weights.

(c) Metric choice changes the conclusion. With 0.17% fraud, a model that flags nothing scores 99.8% accuracy. Papers 6 and 9 show accuracy is misleading; Paper 9 recommends PR-AUC, and Paper 8 adds MCC. Precision, recall, F1, PR-AUC and MCC will be the primary metrics, with accuracy reported only for comparison with earlier work.

(d) One benchmark dominates. Papers 4-8, 13 and 15 all use the same Kaggle credit card dataset. This makes results comparable, but Paper 3 notes that it limits generalisation. The PCA features also limit explanation: V14, V12, V10 and V17 are the top predictors (Paper 15) but have no business meaning. None of the papers report the duplicate rows found in Section 3.4.

(e) Big data and regulation are discussed mainly for the US. Papers 1, 10, 11 and 12 connect detection to Spark pipelines, compliance audit trails, sub-50 ms scoring and customer trust, but in US or general settings. The Irish angle (Central Bank of Ireland fraud statistics, GDPR Article 22, PSD2 Strong Customer Authentication) is still missing from the literature. This is the study's main niche.

2.2 Method comparison matrix

#	Study	Models	Data	Class balance	Reported result	Use here
1*	Jamil et al. 2025	Spark MLlib, Flink, MapReduce + RF	US financial data	n/s	F1 0.89; FP -35%	Big data framing

#	Study	Models	Data	Class balance	Reported result	Use here
2	Elumilade et al. 2025	Conceptual framework	Case studies	n/a	Detection 2.5x faster	Business value
3	Gkegkas et al. 2025	Systematic review (80+)	Mostly Kaggle, IEEE-CIS	Discussed	Ensemble F1 0.82 vs 0.65	Model choice
4*	Salunke et al. 2025	LR, DT, RF	Kaggle ULB	Class weights	RF F1 0.85; LR recall 0.92	Primary baseline
5*	Saeed & Abdulazeez 2024	KNN, RF, LR	Kaggle ULB	n/s	RF AUC 0.98, F1 0.86	Benchmark
6	Appavu 2025	LR, DT	Kaggle ULB	n/s	LR recall 0.89; DT 0.78	Metric warning
7*	Andrade-Arenas & Yactayo-Arias 2025	LR, DT, RF, SVM	Kaggle ULB	SMOTE vs RUS/ROS	RF F1 0.82 to 0.91	Resampling design
8	Aghware et al. 2024	RF + SMOTE / ADASYN	Kaggle ULB	SMOTE	Recall 0.76 to 0.92; MCC	Extra metric
9	Baisholan et al. 2025	Review (50+)	Various	Original imbalance	Ensemble F1 0.78 vs 0.45	PR-AUC choice
10	Chowdhury et al. 2025	ML + compliance checks	Synthetic banking	n/s	94% fraud detected	Regulation
11	Khan et al. 2026	RF, XGBoost, LSTM	FinTech transactions	n/s	RF F1 0.84; < 50 ms	Real-time angle
12	Mensa & Akinsanya 2026	Mixed methods	US survey (500+)	n/a	Trust +28%	Trust / ROI
13	Haider et al. 2024	PCA + SMOTE + LR, DT, RF, SVM	Kaggle ULB	SMOTE	RF F1 0.93, AUC 0.99	Upper reference
14	Bian 2025	Review: ML, ensembles, DL	Various	Discussed	DL +2-5% F1 only	Interpretability
15*	Afriyie et al. 2023	DT, RF, LR, NB, SVM	Kaggle ULB	Stratified k-fold	RF F1 0.86; V14 top	Feature check

* baseline paper. n/s = not stated; n/a = not a classification experiment.

2.3 Refined research gaps and link to research questions

Gap from the synthesis	How this study responds	RQ
G1. SMOTE timing is unreported, so gains may be leakage	Resample the training fold only; compare with class weights on the same split	RQ1
G2. Accuracy-led reporting hides poor fraud detection	Precision, recall, F1, PR-AUC and MCC as primary metrics	RQ1
G3. Duplicate rows in the benchmark are never discussed	Report results with duplicates removed before the split (Section 3.4)	RQ1, RQ3
G4. PCA features give little explanation to regulators	Feature importance plus a discussion of GDPR Article 22 explanation needs	RQ2
G5. No Irish grounding of big data fraud detection	Use Central Bank of Ireland statistics as context; Irish recommendations	RQ2, RQ3

3. Dataset Collection

3.1 Selection criteria and candidates

Five criteria were set before choosing: (1) labelled card or payment transactions, (2) an open licence for academic use, (3) European origin where possible, (4) no personal data, and (5) comparability with the baseline papers. Four candidates were assessed.

Candidate	Type	Size	Fraud rate	Licence	Decision
Credit Card Fraud Detection (ULB, Kaggle)	Real European card transactions, Sept 2013	284,807 rows	0.173%	DbCL-1.0	Selected: named in the proposal, used by 7 of 15 papers, European, anonymised
IEEE-CIS Fraud Detection (Kaggle)	Real e-commerce transactions	about 590k rows	about 3.5%	Competition rules	Rejected: competition-only licence, many raw identity fields
PaySim (Lopez-Rojas et al., 2016)	Simulated mobile money	6.36M rows	about 0.13%	CC BY-SA 4.0	Reserve: synthetic, not card payments
Bank Account Fraud (Jesus et al., 2022)	Synthetic account applications	1M rows per variant	about 1%	CC BY-NC-SA 4.0	Reserve: application fraud, not transactions

For Irish context the Central Bank of Ireland payment fraud statistics (2022-2024) were also collected. They are aggregate figures, used in the introduction and discussion, not for model training: EUR 160 million of fraudulent payments and about 815,000 fraudulent transactions in 2024 (centralbank.ie, 2025).

3.2 Acquisition and verification

The dataset was downloaded with the Kaggle API: `kaggle datasets download -d mlg-ulb/creditcardfraud --unzip -p project/data`. Kaggle metadata lists the licence as DbCL-1.0 (Database Contents License). The file `creditcard.csv` is 150.8 MB. Its SHA-256 checksum is recorded so later weeks can prove the same file is used:

76274b691b16a6c49d3f159c883398e03ccd6d1ee12d9d8ee38f4b4b98551a89

3.3 Data dictionary

V1-V28 are principal components produced by the data owners to protect cardholder confidentiality; the original features are not released. Only Time and Amount keep their original meaning. The full 31-column dictionary is in `data_dictionary.csv`; the table below shows the non-PCA columns and the V1-V28 group.

Column	Type	Missing	Min	Max	Mean	Meaning
Time	float64	0	0.0	172792.0	94813.86	Seconds since the first transaction in the file
Amount	float64	0	0.0	25691.16	88.35	Transaction amount (EUR, not scaled)
Class	int64	0	0.0	1.0	0.002	Target label: 1 = fraud, 0 = legitimate
V1-V28	float64	0	-113.74	120.59	0 (centred)	PCA components; std from 1.959 (V1) down to 0.33

3.4 Initial quality audit

Check	Result	Implication
Rows x columns	284,807 x 31	Fits in memory; all three models train in minutes
Missing cells	0	No imputation needed
Exact duplicate rows	1,081 (19 of them fraud)	Remove before splitting, or copies land in both train and test

Check	Result	Implication
After de-duplication	283,253 legitimate, 473 fraud	This becomes the modelling set
Time span	0 to 48.0 hours (two days)	Too short for a month-level time split
Zero-amount transactions	1,825	Keep; card checks are a known fraud pattern
Amount range	0 to 25,691.16; not scaled	Scale Amount (and Time) like V1-V28
V1-V28 centring	max mean = 0.0	Already centred; no further scaling needed

3.5 Class distribution

Only 492 of 284,807 transactions (0.173%) are fraud: one fraud per 578 legitimate transactions. A model that predicts 'legitimate' every time would reach 99.83% accuracy, which confirms gap G2. The split must be stratified so the test set keeps its fraud cases.

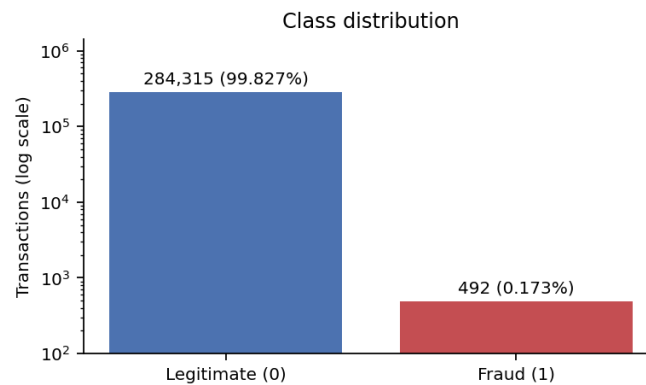


Figure 1. Class distribution (log scale).

3.6 First look at the data

Fraud amounts are skewed toward small values: the median fraud is 9.25 against 22.0 for legitimate transactions, although the fraud mean is higher (122.21 against 88.29) because of a long tail. The largest fraud is 2,125.87. This fits small 'test' charges followed by larger ones (Figure 2).

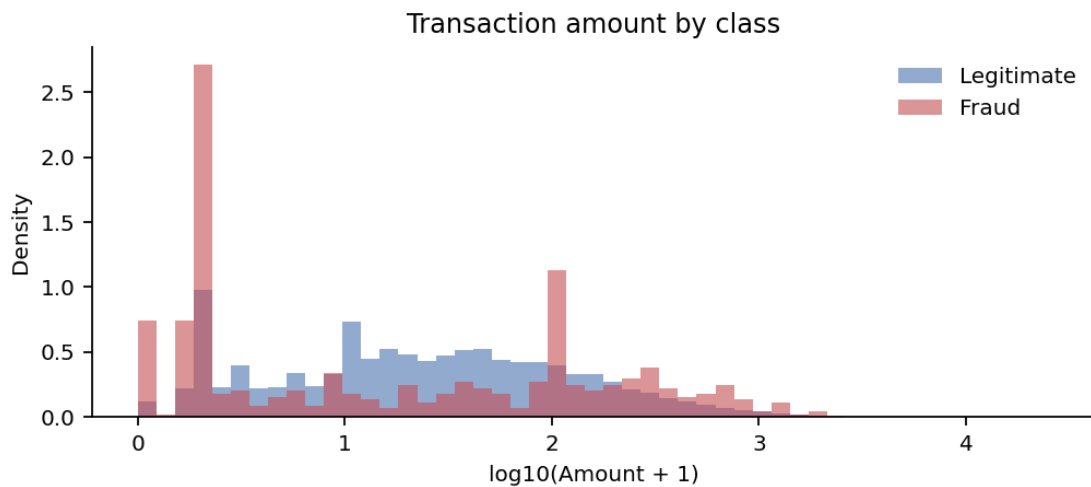


Figure 2. Distribution of $\log_{10}(\text{Amount} + 1)$ by class.

Transactions follow a day-night cycle (Figure 3). Volume drops in the first hours of each day, while the fraud rate rises: 0.518% in hours 0-5 against 0.141% for the rest of the day. An hour-of-day feature derived from Time is therefore worth testing in preprocessing.

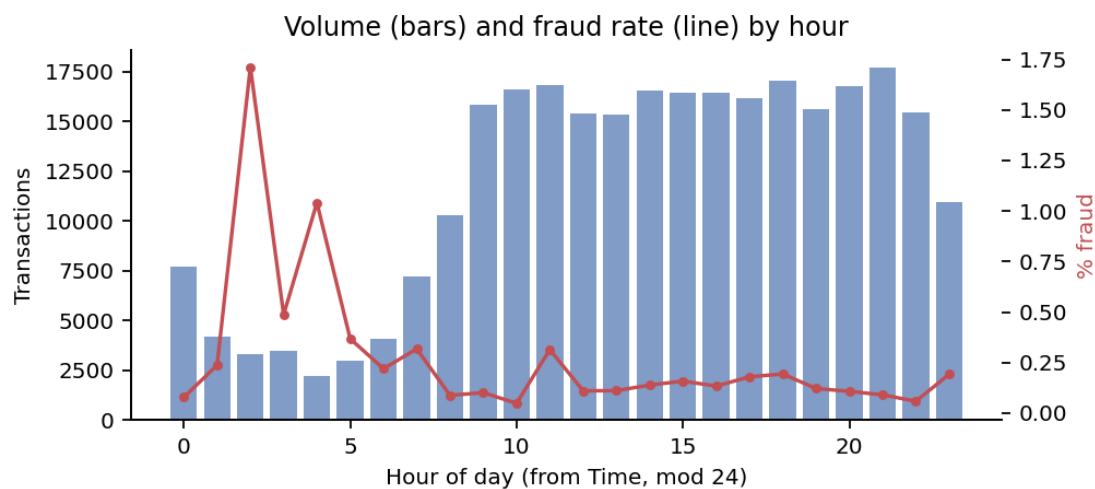


Figure 3. Transactions per hour of day (bars) and fraud rate (line).

The PCA components most correlated with fraud are V17 (0.326), V14 (0.303), V12 (0.261), V10 (0.217), V16 (0.197), V3 (0.193). This matches the V14/V12/V10/V17 finding of Paper 15, so the downloaded file behaves like the one used in the baseline literature (Figure 4).

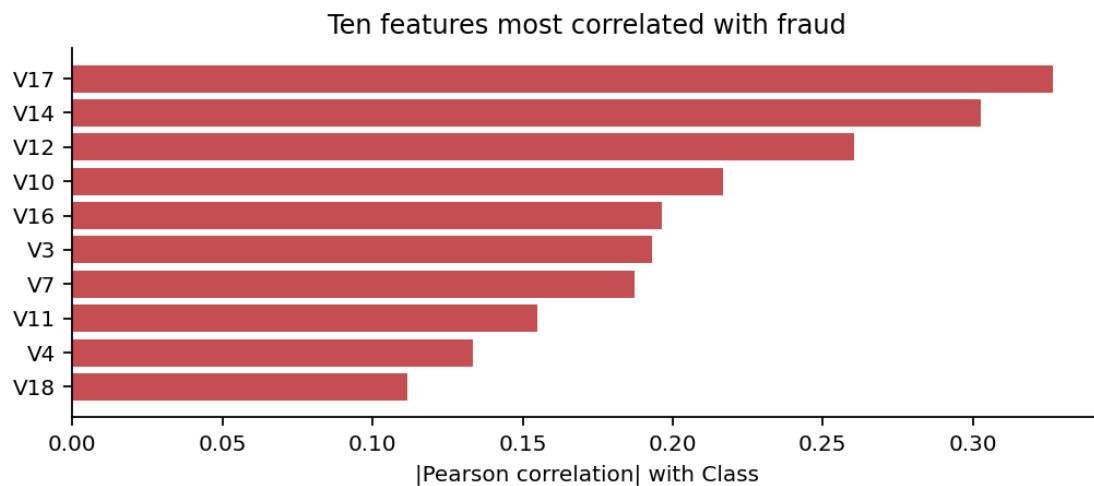


Figure 4. Absolute correlation of each feature with Class (top 10).

Issues logged for Data Preprocessing (W3-W4)

#	Issue	Planned action
1	1,081 duplicate rows	Drop before the train-test split
2	Time and Amount are unscaled	StandardScaler (or RobustScaler for Amount); fit on training data only
3	Time is seconds from start	Derive hour of day; test whether it helps
4	578:1 imbalance	Stratified 70/30 split (as in config.py); class weights vs SMOTE on training fold
5	PCA features lack meaning	Report feature importance; discuss explanation limits

4. Ethics and Data Governance

The dataset contains no names, card numbers or other direct identifiers. The data owners released only PCA components, Time and Amount to protect cardholder confidentiality, so individuals cannot be re-identified from the file. It is used under the DbCL-1.0 licence for academic research, stored locally, not redistributed, and identified by its checksum. The Central Bank of Ireland statistics are public aggregate figures. Because the transactions are European but not Irish and date from 2013, results will be presented as evidence about methods, with Irish recommendations drawn from the literature and Central Bank data rather than claimed from the model.

5. Progress, Risks and Week 3 Plan

Item	Status / note
Literature synthesis, comparison matrix, refined gaps	Done (Section 2)
Dataset selected against written criteria	Done (Section 3.1)
Dataset downloaded, checksummed, licence checked	Done (Section 3.2)
Data dictionary and quality audit	Done; saved as data_dictionary.csv and dataset_profile.json
Irish context data collected	Done: Central Bank of Ireland 2022-2024 fraud statistics
Risk: 2013 data may not reflect current fraud	Stated as a limitation; cite recent Irish statistics
Risk: PCA features limit interpretability	Feature importance plus a regulatory discussion

Week 3 (Literature Review and Dataset Collection finish; Data Preprocessing starts): add 3-5 papers on resampling leakage and PR-AUC evaluation, and on the Irish/EU regulation of automated fraud decisions; close

dataset collection by recording the final de-duplicated modelling set; start the preprocessing notebook (drop duplicates, scale Time and Amount, create the stratified split with a fixed seed and save it for all later experiments).

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